

04-630: Data Structures and Algorithms for Engineers

Assignment III: Text Analysis Using Binary Search Trees

**Carnegie
Mellon
University**
Africa

Rachel Tuyishimire

Rtuyishi

Reflection Report

This project involved the implementation of a Binary Search Tree (BST) to analyze word frequencies in text files. Effective memory management was a key consideration, as each unique word required the allocation of a new node. To optimize memory usage and prevent leaks, the BST was deleted upon completing the processing of each file. Words were stored using fixed-size character arrays to reduce overhead.

BST operations, including insertions and lookups, operate in $O(h)$ time, where h represents the tree height. Ideally, in a balanced BST, this complexity is $O(\log n)$ however, in the worst case of a skewed tree, it degrades to $O(n)$. Since no balancing mechanisms were enforced, performance could decline with sorted input. While AVL trees were considered for self-balancing capabilities, they were excluded to maintain simplicity.

The BST not only stored words and their frequencies but also recorded the depth of each node to analyze operational efficiency. The depth metric facilitated the measurement of probes per operation, particularly during insertions.

Testing was conducted using a variety of input cases, including repetitive words like “love you love dsa must must love dsa must”, single-line sentences, and texts with extensive punctuation. Case insensitivity was verified with variations in capitalization, and mixed alphanumeric inputs were examined. The accuracy of word frequencies, levels, and probe counts was systematically validated. Challenges such as punctuation handling were addressed using `strtok`, while memory deallocation was ensured through a recursive `deleteTree` function.

A hash table was considered as an alternative due to its $O(1)$ lookup time but was ultimately discarded in favor of the BST's ability to maintain sorted order for in-order traversal. The final implementation balanced efficiency, accuracy, and memory management, ensuring reliable text analysis.